

Characterizing Protein Interaction Networks using Graph Representation Learning

Bachelor Thesis



Characterizing Protein Interaction Networks using Graph Representation Learning

Bachelor Thesis
June, 2026

By
Tobias Rodrigues Bjerre
William Kock-Andersen

Copyright: Reproduction of this publication in whole or in part must include the customary bibliographic citation, including author attribution, report title, etc.

Cover photo: Vibeke Hempler, 2012

Published by: DTU, Department of Applied Mathematics and Computer Science,
Brovej, Building 118, 2800 Kgs. Lyngby Denmark
www.compute.dtu.dk

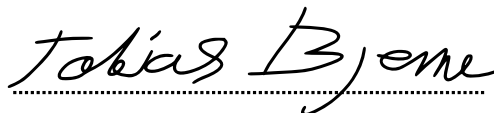
Approval

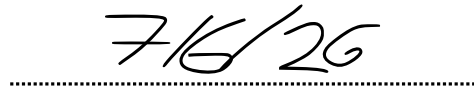
This thesis has been prepared over four months at DTU Compute, Technical University of Denmark, in partial fulfilment for the degree Bachelor of Science in Engineering, BSc Eng.

It is assumed that the reader has basic knowledge of probability, statistics, machine learning, and graph-based representation learning.

The work presented in this thesis has been carried out collaboratively by both authors, who have contributed equally to all parts of the project and share responsibility for the content of the report.

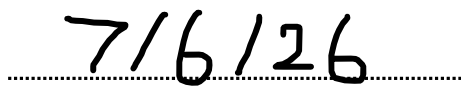
Tobias Rodrigues Bjerre - s234823


.....
Signature


.....
Date

William Kock-Andersen - s234803


.....
Signature


.....
Date

Abstract

Protein–protein interaction (PPI) networks are commonly represented at the gene or canonical protein level, which can obscure important differences between protein isoforms produced by the same gene. Since isoform-specific interactions remain sparse and difficult to measure experimentally, this thesis investigates whether graph representation learning can characterize and predict interactions in an inferred isoform–isoform interaction network.

We evaluate latent distance models (LDMs) for isoform-level link prediction in both transductive and inductive settings. First, a LDM is used to predict held-out interactions between isoforms observed during training. We then introduce a multimodal latent distance model (MMLDM), which uses frozen ESM C sequence embeddings together with gene–isoform membership relations and gene–gene interaction data to predict interactions involving unseen isoforms.

The transductive LDM achieves strong performance, reaching an Average Precision (AP) of 0.795 and ROC-AUC of 0.987. While latent geometry significantly improves performance, a random-effects baseline demonstrates that much of the predictive signal is explained by node-specific interaction tendencies. In the inductive setting, the best MMLDM achieves an AP of 0.498 on unseen isoforms, while gene–isoform and gene–gene modalities provide only marginal improvements. Finally, gene-level structure can be recovered above random baseline performance even without direct gene–gene supervision.

Overall, the results show that latent distance models can effectively characterize inferred isoform interaction networks and that sequence-derived representations support limited generalization to unseen isoforms. More broadly, the results suggest that transferring information from isoform interactions to coarser gene-level organization is easier than using gene-level information to improve isoform-specific prediction, highlighting an important challenge for future graph-based models of protein interaction networks.

Acknowledgements

We would like to express our sincere gratitude to our primary supervisors, Morten Mørup and Kristoffer Vitting-Seerup, for their invaluable guidance, constructive feedback, and continuous support throughout this project. Their expertise and insights were instrumental in shaping the direction and quality of our work.

We also wish to thank Isabella Østerlund for her assistance in providing and interpreting the data, and Oscar Lauritz Møberg for sharing his expertise and offering valuable theoretical insights that contributed significantly to this project.

We are deeply grateful for their support and contributions. Without their assistance, this project would not have reached its present form.

Contents

Approval	ii
Abstract	iii
1 Introduction	1
1.1 Motivation	1
1.2 Approach	2
1.3 Research questions	3
1.4 Thesis structure	4
2 Data	5
2.1 Isoform–isoform interaction graph	5
2.2 Gene–isoform membership graph	7
2.3 STRING-derived gene–gene graph	8
2.4 ESM C sequence features	9
3 Method	11
3.1 Transductive learning setup	11
3.2 Latent distance model	11
3.3 Inductive learning setup	13
3.4 Multimodal latent distance model	13
3.5 Experimental setup	16
3.6 Evaluation metrics	17
4 Results	19
4.1 Transductive isoform–isoform prediction	19
4.2 Inductive isoform–isoform prediction	20
4.3 Gene–gene reconstruction without direct supervision	22
5 Discussion	25
5.1 Node-specific signal and latent geometry	25
5.2 Inductive generalization and auxiliary modalities	26
5.3 Gene-level reconstruction and the direction of transfer	28
5.4 Limitations and future work	29
6 Conclusion	31
Bibliography	33

1 Introduction

1.1 Motivation

PPI networks provide a framework for studying how gene products interact to carry out cellular functions. In these networks, proteins are represented as nodes and their physical or functional interactions as edges, allowing molecular relationships to be analyzed as a graph-structured problem. This representation has become central in systems biology, where interaction networks are used to study cellular organization, disease mechanisms, and potential therapeutic targets [1].

A central limitation of many PPI networks, however, is that they describe interactions at a relatively coarse biological resolution. Large-scale PPI resources are often represented at the level of genes or canonical protein products, which makes interaction data easier to collect, represent, and analyze. This simplification is useful at scale, but it can obscure important biological variation, because a single gene can give rise to multiple protein isoforms through mechanisms such as alternative splicing [2, 3, 4]. As a result, canonical protein-level networks can collapse biologically distinct protein isoforms into a single node, hiding interaction patterns that only become visible at isoform resolution.

Experimental studies have shown that this loss of resolution can be substantial. Isoforms encoded by the same gene can differ in their interaction partners, expression patterns, and functional roles [3]. Isoform-level interactions are therefore not merely a finer annotation of canonical PPI networks, but can change both the local structure and biological interpretation of the network. At the same time, isoform-specific interactions remain difficult to measure experimentally at large scale, and available isoform-level interaction maps are far less complete than canonical PPI resources [3, 4].

Prior work has approached this challenge by constructing and predicting isoform-resolved interactions from indirect biological evidence. The IIDB database collected isoform–isoform and domain–domain interaction information to support analysis at isoform resolution [2], while Ghadie et al. proposed DIIP, a domain-based method for predicting human isoform interactions from reference interactomes and domain–domain interaction evidence [4]. These efforts show that isoform-level interaction modeling is possible, but also highlight the indirect and incomplete nature of currently available isoform interaction data.

This thesis studies isoform-level interaction modeling from a graph representation learning perspective. Using data provided by the Isoform Analysis Group at the Technical University of Denmark (DTU), we analyze a computationally derived isoform–isoform interaction dataset based on canonical PPI data. The labels should therefore be interpreted as inferred rather than as direct experimental observations. Our goal is to investigate whether a latent distance model (LDM) can recover interaction patterns in this sparse inferred network, and whether additional biological modalities can improve prediction for isoforms whose interaction patterns are not directly observed during training.

The project is primarily a machine-learning study. Therefore, any biological interpretations and conclusions should be viewed in the context of machine learning methodologies and are not intended as authoritative biological assessments.

1.2 Approach

The LDM is a classical probabilistic framework for geometric network representation introduced by Hoff et al. [5]. In this model, each node is assigned an unobserved position in a low-dimensional Euclidean space, such that the probability of interaction decreases with the distance between two nodes. This gives the model a direct geometric interpretation, where nearby nodes are expected to have similar relational patterns. Hoff later extended this framework by combining latent interaction terms with node-specific effects, allowing variation in node connectivity to be modeled separately from dyadic latent structure [6]. This distinction is important in sparse biological networks, where some interactions may be predictable from node-level connectivity alone.

Recent work has renewed interest in distance-based graph representations by improving scalability while preserving the interpretability of low-dimensional geometry. For example, Nakis et al. [7] propose the Hierarchical Block Distance Model, which approximates latent distance structure through a multiscale hierarchy while retaining compact geometric embeddings. This work is especially relevant here because it supports the broader premise that distance-based embeddings can remain useful for link prediction and network reconstruction while being easier to interpret than high-dimensional black-box representations.

We first study isoform-level link prediction in a transductive setting, where all isoforms are present during training and the task is to predict held-out interactions between already observed isoforms. This setting fits the LDM, since each isoform can be assigned its own learned latent position and interactions can be modeled through distances in this space. It also provides a useful starting point for evaluating whether the inferred isoform interaction graph contains geometric structure beyond simple node-level connectivity patterns. However, because the resulting embeddings are tied to the nodes observed during training, this model cannot make predictions for isoforms that were absent from the training graph.

This limitation motivates an inductive extension, where latent positions are computed from transferable biological information rather than learned only as node-specific parameters. This follows the general principle behind inductive graph representation learning, where embeddings are generated from node features or local neighborhood structure, as formalized by GraphSAGE [8]. Building on this idea, we extend the LDM to combine sequence-derived isoform information with gene-level relational context, resulting in a multimodal latent distance model (MMLDM).

In the MMLDM, each isoform is represented using a frozen ESM Cambrian (ESM C) embedding, a pretrained protein-language-model representation derived from its amino-acid sequence [9]. These embeddings are used to parameterize both isoform latent positions and isoform-specific random effects, so that predictions for isoforms unseen in the isoform-isoform training graph can depend on sequence-derived information. Gene-isoform and gene-gene relations are used to provide additional biological context beyond the isoform sequence alone, following the design of related work that shows the value of bipartite gene-isoform modeling in the prediction of isoform function [10]. The model therefore reflects the assumption that isoform-level interaction prediction may benefit from combining amino-acid sequence information with broader canonical-level network structure.

Together, these two settings frame the thesis. The transductive setting tests whether latent distance models capture structure in the observed inferred isoform interaction graph beyond node-level connectivity. The inductive setting tests whether sequence-derived isoform features and gene-level relational context can support prediction for isoforms whose

interaction patterns are not directly observed during training. This leads to three research questions that guide the remainder of the thesis.

1.3 Research questions

1. **To what extent can a latent distance model support transductive link prediction in partially observed isoform-level protein interaction networks, and how much of its predictive performance is driven by node degree rather than dyadic latent structure?**

To investigate this, we evaluate an LDM with node-specific random effects on the isoform–isoform interaction graph under a transductive link prediction setting. The data is split at the gene-pair level, preventing highly related isoform pairs from being distributed across splits, and thereby reducing information leakage. In this setting, all isoforms remain observable during training, while a subset of interaction labels is withheld for evaluation. To assess the contribution of latent geometric structure beyond node-level connectivity, we compare models with latent dimensions $d \in \{2, 8, 32\}$ against a random-effects-only baseline ($d = 0$). This allows us to examine both whether latent distance structure improves predictive performance and how performance changes as the capacity of the latent space increases.

2. **How well can a multimodal latent distance model integrating sequence-derived isoform features, gene–isoform structure, and gene–gene interactions support inductive link prediction on unseen isoforms, and what is the contribution of each data source?**

To investigate this, we evaluate the MMLDM under a gene-level inductive split. Genes are assigned to training, validation, and test partitions, and the model is trained on isoform–isoform labels only for pairs where both genes belong to the training split. Validation and test pairs therefore always contain at least one isoform whose target interaction labels are not observed during training. To make prediction possible in this setting, isoform latent positions and random effects are computed from frozen ESM C embeddings rather than learned as free node-specific parameters. We compare models with latent dimensions $d \in \{2, 8, 32\}$ against a sequence-derived random-effects-only baseline ($d = 0$). To assess the contribution of additional biological context, we evaluate three configurations: isoform–isoform interactions alone (G_{II}), isoform–isoform interactions combined with gene–isoform relations ($G_{II} + G_{GI}$), and a full multimodal configuration additionally incorporating gene–gene interactions ($G_{II} + G_{GI} + G_{GG}$).

3. **How accurately can a model trained without direct gene–gene supervision recover STRING gene–gene interactions, and what does this reveal about the relationship between isoform-level interactions and canonical gene-level organization?**

Finally, we investigate whether gene-level structure emerges when the MMLDM is trained without direct G_{GG} supervision. For this experiment, we use the $G_{II} + G_{GI}$ configuration and exclude the STRING-derived gene–gene graph from the training objective. We then evaluate whether STRING gene–gene interactions can be recovered from the trained model using two readouts: a distance readout based on the learned gene positions u_g , and an aggregation readout based on the strongest predicted isoform–isoform score across isoform pairs spanning each candidate gene pair. This tests whether isoform-level interactions and gene–isoform membership induce gene-level organization that aligns with canonical STRING structure, and

whether this alignment is captured more clearly by compressed gene embeddings or by aggregated isoform-level predictions.

1.4 Thesis structure

The remainder of the thesis is structured as follows. Chapter 2 introduces the datasets and feature representations used to construct the graph-based modeling problem. Chapter 3 presents the transductive and inductive modeling setups, the latent distance models used in each setting, and the experimental setup and evaluation metrics. Chapter 4 reports the performance of the models across the three research questions. Chapter 5 interprets the results in relation to the structure of the data, the role of auxiliary biological information, and the limitations of the approach. Finally, Chapter 6 summarizes the main findings of the thesis.

2 Data

The data used in this project were provided by the Isoform Analysis Group at DTU and are organized around a central isoform–isoform dataset. Each record describes a pair of isoforms through protein identifiers (`ensp_1`, `ensp_2`) and gene identifiers (`gene_1`, `gene_2`). For each pair, a Bayesian inference framework based on physical interaction evidence from the STRING database [11] assigns an interaction probability $\pi \in [0, 1]$. The resulting records should therefore be interpreted as inferred isoform-level interaction labels rather than direct experimental observations.

Each record additionally contains a binary interaction label `interact`, which is obtained by thresholding the interaction probability at 0.5: $\pi \geq 0.5 \Rightarrow 1$, otherwise 0. The dataset contains 2,865,291 interaction records spanning 58,578 unique isoforms. Figure 2.1 shows the distribution of π .

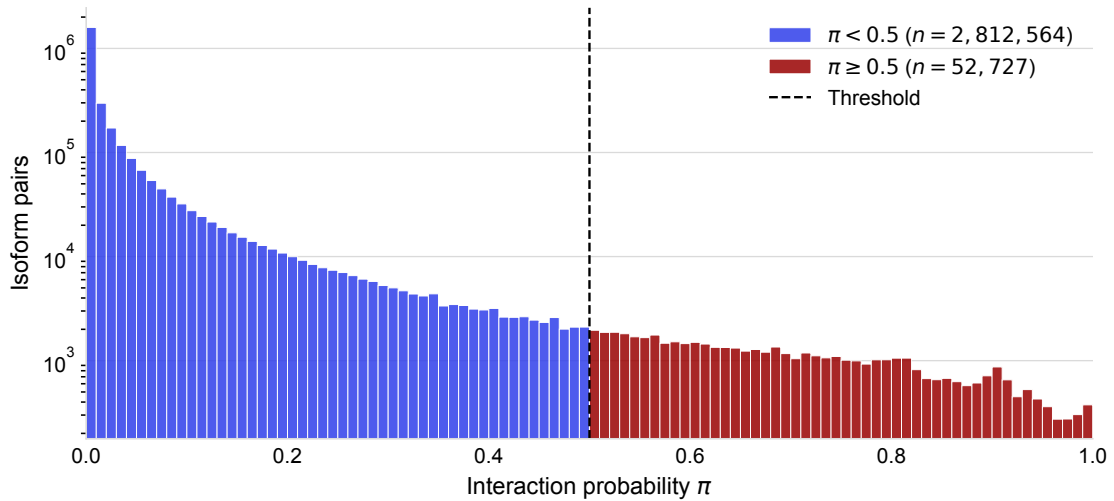


Figure 2.1: Histogram of π with log-scaled count on the y-axis

As shown in Figure 2.1, the interaction probabilities are highly skewed towards small values with $\text{median}(\pi) = 0.006$, and only a small fraction of observations concentrated in the upper tail. After thresholding, only 1.84% of the interaction records are positive, resulting in a highly imbalanced prediction setting.

The probability values π are not used directly for modeling, since the project focuses on unweighted graphs and binary link prediction. All subsequent analysis therefore uses the thresholded labels. This binarization introduces an important limitation: pairs just below the threshold are treated as equivalent negatives to pairs with almost no interaction evidence, even though the underlying probabilities may carry very different signals. Similarly, all pairs with $\pi \in [0.5, 1.0]$ are treated as equivalent positives. This loss of information may limit the predictive performance attainable on the binarized network, particularly for isoform pairs close to the threshold.

2.1 Isoform-isoform interaction graph

We first construct the isoform–isoform graph G_{II} from the thresholded interaction labels. Let $N_I = \{1, \dots, n\}$ denote the isoform node set, with $n = 58,578$. Positive labels are

represented as undirected, unweighted edges between isoforms. The graph is encoded by a binary adjacency matrix $Y_{II} \in \{0, 1\}^{n \times n}$, where $y_{ij}^{II} = y_{ji}^{II} = 1$ if isoforms i and j are labeled as interacting, and 0 otherwise.

The upper triangle of Y_{II} contains $\binom{58,578}{2} = 1,715,661,753$ possible isoform pairs, of which the dataset provides 2,865,291 labeled interaction records. Following the closed-world interpretation specified for the dataset, isoform pairs not included among the provided records can be treated as implicit negative examples. This corresponds to more than 1.7 billion additional negative pairs outside the central dataset. In preliminary experiments, we expanded the training set by sampling negative examples from this larger pool. This did not improve predictive performance beyond what was already obtained from the explicit negative labels in the dataset, while substantially increasing computational cost. We therefore restrict training to the provided labeled records.

We construct and analyse the graph in Python using NetworkX [12]. Summary statistics for G_{II} are shown in Table 2.1.

Table 2.1: Summary statistics for the isoform–isoform graph G_{II} .

Statistic	Value
Nodes (isoforms)	58 578
Edges (positive interactions)	52 727
Density	0.000 031
Nodes with no interactions	49 484
Nodes with interactions	9094
Mean degree	1.80
Median degree	0.00
Max degree	376

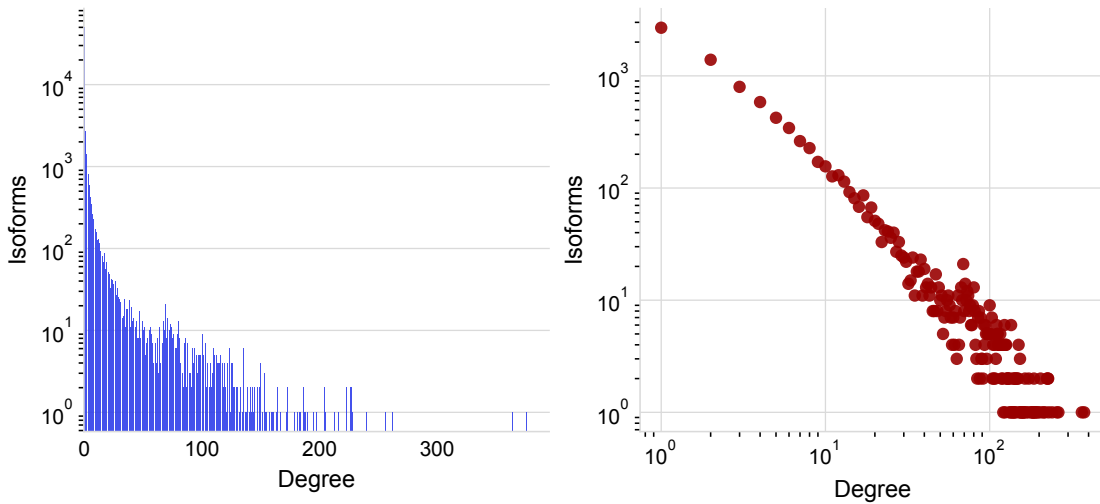


Figure 2.2: Left: degree distribution of G_{II} with log-scaled count. Right: degree distribution restricted to connected isoforms, shown on log-scaled axes.

Together, Table 2.1 and Figure 2.2 highlight the sparsity and degree heterogeneity of G_{II} . A total of 49,484 isoforms, corresponding to 84.5% of all isoforms, have degree zero after thresholding and therefore do not participate in positive interactions. Among the

connected isoforms, most have low degree, while a small number of nodes act as highly connected hubs. This is reflected in the zero median degree, the mean degree of 1.80, the maximum degree of 376, and the long right tail of the degree distribution.

2.2 Gene-isoform membership graph

We next construct the bipartite gene–isoform membership graph G_{GI} , which links each isoform to the gene that codes for it. The isoform node set is the same N_I used in G_{II} . Let $N_G = \{1, \dots, m\}$ denote the set of genes appearing in the isoform dataset, with $m = 10,644$. The graph is represented by a binary membership matrix $Y_{GI} \in \{0, 1\}^{m \times n}$, where $y_{gi}^{GI} = 1$ if isoform i belongs to gene g , and 0 otherwise.

Because each isoform belongs to exactly one gene, every column of Y_{GI} is a one-hot vector encoding gene membership. Consequently, every isoform has degree one in G_{GI} , while gene-side degrees correspond to the number of isoforms associated with each gene. Gene-side degree statistics are reported in Table 2.2.

Table 2.2: Gene-side degree statistics for G_{GI} .

Statistic	Value
Mean degree	5.5
Median degree	4.0
Min degree	1
Max degree	87

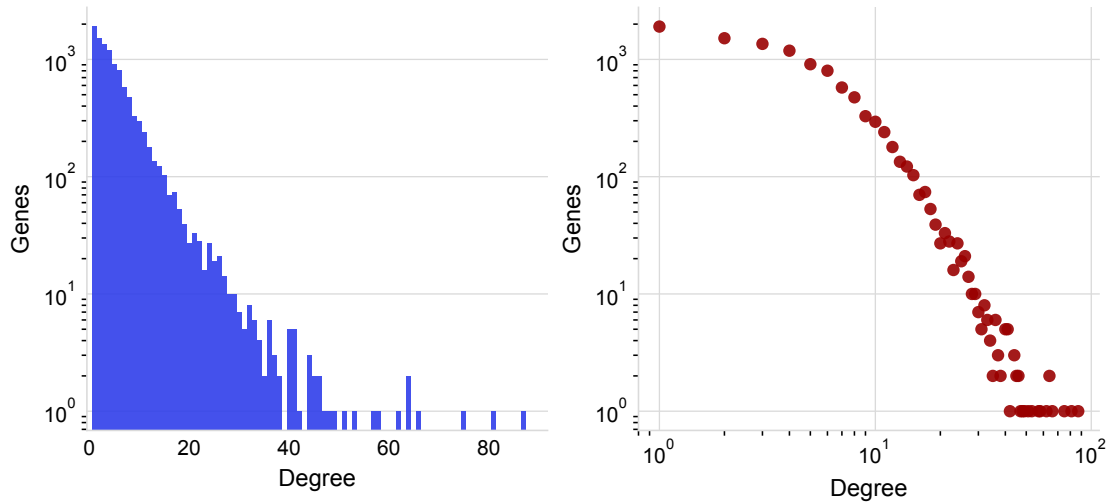


Figure 2.3: Left: gene-membership degree distribution with log-scaled counts. Right: log–log representation of the degree distribution.

Together, Table 2.2 and Figure 2.3 show that most genes are associated with relatively few isoforms, while a smaller number have many. The mean gene-side degree is 5.5, the median is 4, and the largest gene is associated with 87 isoforms, giving a right-skewed degree distribution. Compared with G_{II} , this distribution is less extreme because every isoform contributes exactly one membership edge, although the right tail still indicates substantial variation in the number of isoforms associated with each gene.

2.3 STRING-derived gene-gene graph

Finally, we construct the gene–gene graph G_{GG} from a STRING-derived physical interaction dataset. The interactions are originally defined at the protein-pair level using weighted STRING scores [11], with pairs assigned a positive interaction label when the score is at least 700. To obtain a gene-level graph, protein identifiers are mapped to their corresponding genes, and positive protein-pair interactions are represented as gene–gene edges.

Let $N_{G+} = \{1, \dots, m_+\}$ denote the gene node set used for G_{GG} , with $m_+ = 18,428$. This set contains the genes from N_G together with 7,784 additional genes that appear only in the STRING-derived interaction data. These STRING-only genes do not have isoform membership edges in G_{GI} , but they are retained because they can still shape the gene-level auxiliary space through G_{GG} and through interactions with genes that do appear in the isoform dataset.

The graph is encoded by a binary adjacency matrix $Y_{GG} \in \{0, 1\}^{m_+ \times m_+}$, where $y_{gh}^{GG} = y_{hg}^{GG} = 1$ if genes g and h are connected by at least one positive mapped protein-level interaction, and 0 otherwise. As in G_{II} , the graph is treated as undirected and unweighted after thresholding. Summary statistics for G_{GG} are shown in Table 2.3.

Table 2.3: Summary statistics for G_{GG} .

Statistic	Value
Nodes (genes)	18 428
Edges (positive interactions)	84 044
Density	0.000 495
Nodes with no interactions	7874
Nodes with interactions	10 554
Mean degree	9.12
Median degree	1.00
Max degree	428

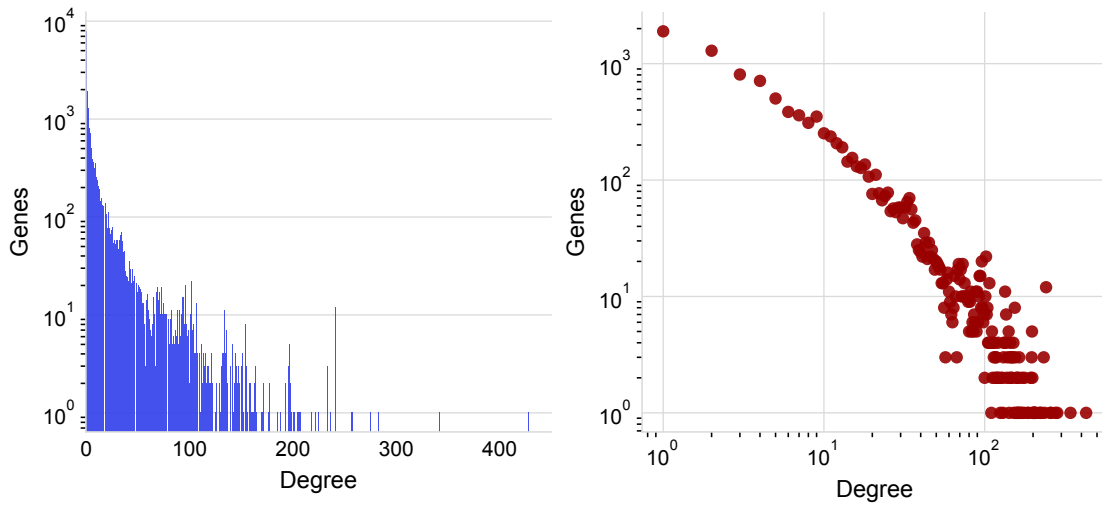


Figure 2.4: Left: degree distribution of G_{GG} with log-scaled counts on the y-axis. Right: log–log representation of the degree distribution.

Together, Table 2.3 and Figure 2.4 show that G_{GG} is more connected than the isoform–isoform graph, but still highly heterogeneous. The graph contains 84,044 positive interactions among 18,428 genes, with a density of 0.000495 and a mean degree of 9.12. At the same time, the median degree is only 1, and 7,874 genes have no positive interactions, while a small number of genes act as highly connected hubs with degrees up to 428. Compared with G_{II} , the STRING-derived gene–gene graph has broader coverage at the canonical level, but its degree distribution remains strongly right-skewed.

2.4 ESM C sequence features

Each isoform is associated with a sequence-derived feature vector produced by ESM C, a protein language model from the ESM Cambrian family [9]. Protein language models represent amino-acid sequences in a learned feature space, where sequence patterns are encoded as numerical embeddings. In this project, each isoform i is represented by a precomputed global embedding $f_i \in \mathbb{R}^{1152}$, generated from its amino-acid sequence. The embeddings are aligned with the isoform node set N_I used in G_{II} and G_{GI} , so that each isoform node has a corresponding sequence feature vector.

The ESM C embeddings are treated as fixed input features and are not fine-tuned during training. In the MMLDM, these features are used to parameterize isoform latent positions and isoform-specific random effects, allowing representations to be computed from sequence rather than learned as free node-specific parameters. Since the feature dimension $D = 1152$ is substantially larger than the latent dimensions d used in the model, a learned projection from $\mathbb{R}^D$ to $\mathbb{R}^d$ is required to integrate the embeddings into the shared latent space. This projection is introduced in Section 3.4.

To assess whether the sequence features provide a clear basis for distinguishing isoforms from the same gene, we compared cosine similarities between ESM C embeddings for isoform pairs from the same gene and a random sample of isoform pairs from different genes. The result is shown in Figure 2.5.

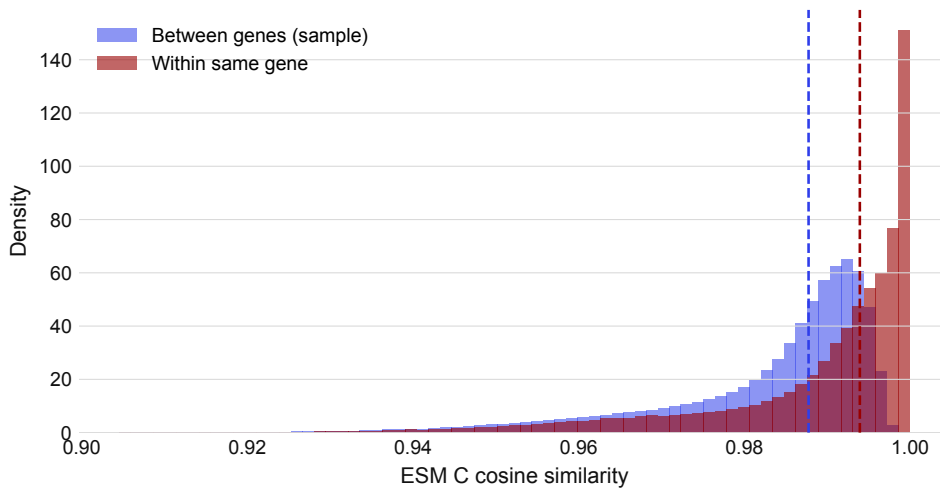


Figure 2.5: Cosine similarity distributions for ESM C embeddings. The within-gene distribution contains all same-gene isoform pairs with available embeddings ($n = 296,174$), while the between-gene distribution is an equally sized random sample of isoform pairs from different genes. Dashed lines mark the medians.

Within-gene isoform pairs are extremely similar in the ESM C feature space, with median cosine similarity 0.994 and interquartile range $[0.985, 0.998]$. Between-gene pairs are also

highly similar, with median 0.988 and interquartile range [0.978, 0.992]. The within-gene distribution is shifted upward, but the two distributions overlap substantially. Together, these distributions indicate that global ESM C embeddings preserve strong sequence similarity within genes, but provide limited contrast for separating isoform-specific interaction behaviour.

3 Method

The primary task addressed in this work is link prediction on the isoform–isoform graph G_{II} . Given a candidate isoform pair (i, j) , the goal is to learn a model that assigns a probability to the event $y_{ij}^{II} = 1$, corresponding to a predicted interaction between the two isoforms.

This chapter introduces the modeling and evaluation framework used for this task. We first define the transductive setting, where the model predicts held-out interaction labels between isoforms that are largely observed during training, and introduce the LDM used in this setting. We then define the inductive setting, where evaluation involves isoforms whose target isoform–isoform interaction labels are absent from training, and introduce the MMLDM, which computes isoform representations from ESM C sequence features and incorporates gene-level biological structure as auxiliary information. Finally, we describe the experimental setup and evaluation metrics used to compare the models across the three research questions.

3.1 Transductive learning setup

In the transductive setting, the model predicts held-out isoform–isoform labels while learning freely parameterized representations for the isoforms observed during training. For G_{II} , the split is performed at the gene-pair level: gene pairs are assigned to training, validation, and test splits, and all isoform–isoform records belonging to the same gene pair are kept in the same split. This reduces leakage between highly related isoform pairs from the same two genes.

Because individual isoforms may participate in interactions across multiple gene pairs, the same isoform can appear across several splits through different interaction partners. As a result, almost all isoforms in N_I remain observable during training, making the task transductive for the overwhelming majority of nodes. A small number of isoforms occur only in validation or test splits. For these isoforms, the freely learned latent positions and random effects receive no direct training signal. This is a minor deviation from the ideal transductive setting, but it preserves the gene-pair split used to reduce leakage.

3.2 Latent distance model

In the transductive setting, we model G_{II} by assigning each isoform a freely learned latent representation. For readability, we omit the II superscript in this section and write y_{ij} for isoform–isoform labels. Let

$$Z = \{z_1, \dots, z_n\}, \quad (3.1)$$

where each isoform $i \in N_I$ has a latent position $z_i \in \mathbb{R}^d$ in a low-dimensional Euclidean space.

LDMs are well suited to interaction networks because they naturally capture two common structural properties: homophily, where similar nodes are more likely to interact, and transitivity, where interactions tend to cluster locally. The LDM assumes that the probability of interaction between two nodes decreases as the Euclidean distance between their latent positions increases. In the formulation of Hoff et al. [5],

$$\text{logit}(P(y_{ij} = 1)) = \alpha + \beta^\top x_{ij} - \|z_i - z_j\|, \quad (3.2)$$

where α and β are learnable parameters and x_{ij} is a dyad-specific covariate vector. In our setting no dyad-specific covariates are used, so $x_{ij} = \mathbf{0}$. α is a learnable global intercept value for the likelihood.

As shown in Section 2.1, G_{II} is sparse and has substantial degree heterogeneity. To account for variation in how often individual isoforms participate in interactions, we use the random-effects extension introduced by Hoff [6]:

$$\text{logit}(P(y_{ij} = 1)) = \alpha + r_i + r_j - \|z_i - z_j\|, \quad (3.3)$$

where r_i and r_j are isoform-specific random effects. Positive random effects increase the log-odds of interactions involving the corresponding isoform independently of its latent position. These terms therefore capture isoform-specific interaction tendency separately from the dyadic latent distance.

Although the node-specific random effects can, in principle, absorb part of the global intercept α , we retain the intercept to separate the global baseline interaction rate from isoform-specific deviations. The random effects capture heterogeneity in node connectivity, while α controls the overall baseline of the model.

We model the interaction network under the assumption that each individual label follows a Bernoulli distribution,

$$y_{ij} \sim \text{Bernoulli}(\hat{\pi}_{ij}), \quad (3.4)$$

where the interaction probability $\hat{\pi}_{ij}$ is obtained by applying the sigmoid function σ to the logit:

$$\hat{\pi}_{ij} = \sigma(\alpha + r_i + r_j - \|z_i - z_j\|). \quad (3.5)$$

This ensures $\hat{\pi}_{ij} \in [0, 1]$. For a single observed label, the Bernoulli likelihood is

$$P(y_{ij} \mid \hat{\pi}_{ij}) = \hat{\pi}_{ij}^{y_{ij}} (1 - \hat{\pi}_{ij})^{1-y_{ij}}. \quad (3.6)$$

Taking the logarithm gives the per-edge log-likelihood:

$$\begin{aligned} \log P(y_{ij} \mid \hat{\pi}_{ij}) &= \log \left(\hat{\pi}_{ij}^{y_{ij}} (1 - \hat{\pi}_{ij})^{1-y_{ij}} \right) \\ &= y_{ij} \log \hat{\pi}_{ij} + (1 - y_{ij}) \log(1 - \hat{\pi}_{ij}). \end{aligned} \quad (3.7)$$

Let $\mathcal{D}_{II}$ denote the observed isoform–isoform records used for training. Maximizing the joint likelihood across these observed interaction labels is equivalent to minimizing the negative log-likelihood

$$L(\hat{\pi} \mid \mathcal{D}_{II}) = - \sum_{(i,j,y_{ij}) \in \mathcal{D}_{II}} [y_{ij} \log \hat{\pi}_{ij} + (1 - y_{ij}) \log(1 - \hat{\pi}_{ij})], \quad (3.8)$$

corresponding to the standard binary cross-entropy loss function.

Because the interaction labels are highly imbalanced, we apply positive-class weighting in the binary cross-entropy objective. This follows the standard weighted BCE formulation used in `BCEWithLogitsLoss`, where the positive term is scaled to adjust the loss for imbalance between negative and positive samples [13]:

$$L(\hat{\pi} \mid \mathcal{D}_{II}) = - \sum_{(i,j,y_{ij}) \in \mathcal{D}_{II}} [p_c y_{ij} \log \hat{\pi}_{ij} + (1 - y_{ij}) \log(1 - \hat{\pi}_{ij})], \quad (3.9)$$

where p_c denotes the ratio of negative to positive interactions in the training set.

3.3 Inductive learning setup

The inductive setting is designed to evaluate whether the model can generalize to isoforms whose target isoform–isoform interactions are not observed during training. Unlike the transductive split, where almost all isoforms remain observable through other interaction partners, the inductive split is performed at the gene level. This means assigning whole genes instead of gene-pair-specific interactions to a split. This makes the evaluation stricter, as holding out a gene also holds out its associated isoforms from the isoform–isoform training graph.

Each gene is assigned to the training, validation, or test split. Isoform–isoform interactions are then assigned according to the genes involved: interactions involving a test gene are assigned to the test split, interactions involving a validation gene but no test gene are assigned to the validation split, and only interactions between two training genes are assigned to the training split. The model is therefore trained exclusively on isoform–isoform labels between training genes.

Consequently, every validation or test interaction contains at least one isoform unseen in the isoform–isoform training graph. These held-out interactions can be divided into one-unseen pairs, where one node is unseen during training, and two-unseen pairs, where both nodes are unseen. We use this distinction in the evaluation to separate interactions involving one training-observed node from the stricter case where both nodes are held out.

The inductive restriction applies only to the target isoform–isoform labels. Held-out isoforms are still represented through auxiliary information such as their amino-acid sequence, gene membership, and gene-level interactions. We treat these sources as priors, since we assume this information is available for all isoforms.

3.4 Multimodal latent distance model

The MMLDM extends the LDM to the inductive setting by computing isoform representations from sequence-derived features rather than learning one free latent vector per isoform. For each isoform i , the precomputed ESM C embedding f_i is mapped into the latent space by a learned projection, producing an isoform position $z_i \in \mathbb{R}^d$. The same sequence features are also used to produce isoform-specific random effects. This allows the model to assign latent positions and random effects to isoforms whose isoform–isoform labels are absent from training.

The model represents isoforms and genes in a shared Euclidean latent space. Isoform positions z_i are sequence-parameterized, while each gene $g \in N_{G+}$ is assigned a learnable latent position $u_g \in \mathbb{R}^d$. The gene–isoform graph G_{GI} and gene–gene graph G_{GG} provide auxiliary training signal in this shared space. All active modalities are modeled with latent-distance likelihoods, so edge probabilities decrease with distance while modality-specific intercepts and node-specific terms account for differences in connectivity.

3.4.1 Isoform-isoform interaction likelihood

Isoform interactions are modeled as

$$\hat{\pi}_{ij}^{II} = \sigma(\alpha_{II} + r_i^{II} + r_j^{II} - \beta_{II}^+ \|z_i - z_j\|). \quad (3.10)$$

This model has the same distance-based form as Equation (3.5), but with two important changes. First, the distance term is scaled by the positive parameter β_{II}^+ , allowing the isoform–isoform modality to learn how strongly latent distance should affect the interaction likelihood. Second, the isoform embeddings z_i and z_j together with isoform-specific

random effects r_i^{II} and r_j^{II} are no longer free node parameters, but are produced from projected ESM C features.

3.4.2 Gene-isoform membership likelihood

The bipartite graph models gene membership:

$$\hat{\pi}_{gi}^{GI} = \sigma(\alpha_{GI} + \gamma_g - \beta_{GI}^+ \|u_g - z_i\|), \quad (3.11)$$

where α_{GI} is a global intercept for the gene–isoform modality, γ_g is a gene-specific intercept acting as a random effect, and β_{GI}^+ is a positive distance scaling parameter. Isoform-specific random effects are not included, since each isoform has exactly one positive edge.

Because the isoform position z_i participates directly in the bipartite likelihood, gradients from gene membership edges influence isoform representations by pulling isoforms towards the positions of their associated genes.

3.4.3 Gene-gene interaction likelihood

Canonical-level gene interactions are modeled similarly to isoform-specific interactions:

$$\hat{\pi}_{gh}^{GG} = \sigma(\alpha_{GG} + r_g^{GG} + r_h^{GG} - \beta_{GG}^+ \|u_g - u_h\|), \quad (3.12)$$

where α_{GG} is a global intercept for the gene–gene modality, and r_g^{GG} and r_h^{GG} are gene-level random effects specific to the gene–gene modality.

Because the gene positions u_g are shared with the bipartite graph, gene–gene interactions indirectly influence isoform representations through the shared latent geometry.

The distance scaling parameters are written as β_{II}^+ , β_{GI}^+ , and β_{GG}^+ because they are constrained to be positive using a softplus reparameterization:

$$\beta^+ = \ln(1 + e^\beta). \quad (3.13)$$

This ensures increasing latent distances decreases the edge probability in each modality.

3.4.4 Sequence-parameterized isoform representations

Isoform positions are not learned as free parameters. Instead, they are produced by a learned projection of the precomputed ESM C feature vector

$$f_i \in \mathbb{R}^D. \quad (3.14)$$

Specifically,

$$z_i = g_\theta(f_i), \quad g_\theta : \mathbb{R}^D \rightarrow \mathbb{R}^d, \quad (3.15)$$

where g_θ is a two-layer multilayer perceptron (MLP) with hidden dimension 42, ReLU activation, and dropout applied to the hidden layer:

$$g_\theta(f) = W_2 \text{ReLU}(\text{Dropout}(W_1 f + b_1)) + b_2. \quad (3.16)$$

The narrow hidden bottleneck encourages the network to extract a compact representation from the much higher-dimensional ESM C embedding space rather than reproducing the original features directly.

A second MLP

$$h_\phi : \mathbb{R}^D \rightarrow \mathbb{R} \quad (3.17)$$

with identical architecture, except for the output dimension, produces the isoform-specific random effect from the same features:

$$r_i^{II} = h_\phi(f_i). \quad (3.18)$$

This replaces the freely learned isoform random effects used in the LDM, ensuring that both latent positions and isoform-specific random effects are determined by sequence information for isoforms without target interaction labels in the training graph.

3.4.5 Joint training objective

Each graph modality is trained using the same weighted binary cross-entropy formulation in Equation (3.9), where positive class weights

$$p_c^{II}, \quad p_c^{GI}, \quad p_c^{GG} \quad (3.19)$$

are set according to the ratio of negative to positive interactions within each training dataset. This compensates for class imbalance separately in the isoform–isoform, gene–isoform, and gene–gene graphs, rather than applying a single global class weight across all modalities.

The construction of negative examples differs between modalities. For G_{II} , we use the labeled isoform–isoform records described in Section 2.1, containing both positive and explicit negative examples after thresholding π . We do not add the full set of implicit non-edges from the complete isoform-pair space. For G_{GI} , positive examples are observed gene–isoform membership edges, while negative examples are sampled from non-membership gene–isoform pairs using a ratio of 10 negatives per positive membership edge. For G_{GG} , positives and negatives are taken from the STRING-derived gene-pair data after thresholding, without adding additional implicit gene–gene non-edges. The positive-class weights p_c^{II} , p_c^{GI} , and p_c^{GG} are computed from these constructed training sets.

Let $\mathcal{M} \subseteq \{II, GI, GG\}$ denote the set of active modalities in a given experiment. The weighted binary cross-entropy loss is computed as the mean loss over the corresponding mini-batch. The total loss for one optimization step is a weighted sum of the active modality-specific losses:

$$\mathcal{L} = \sum_{m \in \mathcal{M}} \tilde{\lambda}_m \mathcal{L}_m. \quad (3.20)$$

where $\mathcal{L}_{II}$, $\mathcal{L}_{GI}$, and $\mathcal{L}_{GG}$ denote the mini-batch losses for the three graph modalities. The effective weights $\tilde{\lambda}_{II}$, $\tilde{\lambda}_{GI}$, and $\tilde{\lambda}_{GG}$ are derived from specified modality weights

$$\lambda_{II}, \quad \lambda_{GI}, \quad \lambda_{GG}, \quad (3.21)$$

which control the intended relative contribution of each modality. The three modalities are optimized jointly. At each optimization step, one mini-batch is drawn from each active graph dataset, and the three losses are combined into a single backward pass. This means that gradients from isoform–isoform interactions, gene–isoform membership, and gene–gene interactions jointly shape the shared latent space.

Because the three datasets contain different numbers of mini-batches, each epoch is defined by the longest dataloader. Let

$$S = \max(S_{II}, S_{GI}, S_{GG}), \quad (3.22)$$

where S_m is the number of mini-batches in modality m . Shorter dataloaders are cycled so that each optimization step contains a batch from every active modality. To prevent a shorter dataloader from being over-weighted simply because its batches are repeated, each modality weight is divided by its cycling factor:

$$\tilde{\lambda}_m = \frac{\lambda_m}{S/S_m}, \quad m \in \{II, GI, GG\}. \quad (3.23)$$

Repeated cycling does not by itself increase the total contribution of a smaller modality within an epoch. Instead, the relative importance of the modalities is controlled by the specified λ values after correcting for dataloader length.

In the experiments, isoform–isoform link prediction is the primary task, while the gene–isoform and gene–gene graphs are used as auxiliary sources of biological structure. We therefore assign the largest weight to the isoform–isoform objective and smaller weights to the auxiliary objectives:

$$\lambda_{II} = 1.0, \quad \lambda_{GI} = 0.5, \quad \lambda_{GG} = 0.25. \quad (3.24)$$

In the inductive experiments, the auxiliary gene–isoform and gene–gene graphs are not evaluated as held-out target tasks. When these modalities are active, their constructed training pairs may include held-out isoforms, because they are treated as prior information for the shared latent space rather than as target labels for the inductive evaluation. Validation and test metrics are computed only on the held-out isoform–isoform interactions defined in Section 3.3.

3.5 Experimental setup

In the transductive experiments, LDMs are trained only on G_{II} with latent dimensions $d \in \{0, 2, 8, 32\}$. The $d = 0$ model has no latent positions and serves as a random-effects-only baseline. Comparing the remaining dimensions against this baseline measures the added value of latent geometry when isoforms have freely learned parameters.

The inductive experiments vary the same four latent dimensions across three modality configurations: G_{II} , $G_{II} + G_{GI}$, and $G_{II} + G_{GI} + G_{GG}$. These correspond to isoform–isoform supervision alone, the addition of gene–isoform membership, and the further addition of STRING-derived gene–gene interactions. This gives $3 \times 4 = 12$ inductive configurations.

The $G_{II} + G_{GI}$ configuration is also used for gene–gene reconstruction without direct G_{GG} supervision. In this case, any recovery of STRING gene–gene edges must come from G_{II} , G_{GI} , and the shared latent space. The specific gene network we reconstruct is not the full G_{GG} , but instead the reduced version consisting of genes in N_G . We therefore only evaluate gene embeddings that are present in the isoform–isoform network, since the external genes are meaningless to evaluate in this setting. Since α_{GG} , r_g^{GG} , r_h^{GG} , and β_{GG}^+ receive no gradients when G_{GG} is excluded, candidate gene pairs cannot be scored with the full likelihood in Equation (3.12). We therefore use a distance readout,

$$s_{gh}^{GG} = -\|u_g - u_h\|. \quad (3.25)$$

This differs from Equation (3.12) in that the global intercept and both node-specific random effects are set to zero, and the distance scaling is fixed to unity, since none of these terms are observed in this configuration.

As a second readout, each candidate gene pair is scored by the maximum isoform–isoform logit over all isoform pairs spanning the two genes. The distance readout evaluates the learned gene embeddings directly. The aggregation readout asks whether the strongest predicted isoform-level interaction recovers the gene-level edge.

Random data splitting and model initialization can affect performance across experiments. All experiments are repeated across 10 random seeds (42–51), with each seed controlling both the data split and model initialization. This produces matched performance values for statistical comparisons across configurations, enabling a more robust assessment of model generalization.

For both transductive and inductive experiments, data were split into 70% training, 10% validation, and 20% test sets. Models were optimized with Adam [14] using a learning rate of 10^{-3} and weight decay of 10^{-5} . We used PyTorch’s `ReduceLROnPlateau` scheduler [15] with factor 0.5 and patience 2, monitoring validation AP. Each model was trained for at most 20 epochs, with early stopping after 5 epochs without improvement in validation AP.

The implementation used for preprocessing, model training, evaluation, and figure generation is available in the accompanying project repository [16].

3.6 Evaluation metrics

We evaluate the models on link prediction performance. Because the interaction data is highly imbalanced, threshold-independent ranking metrics are especially relevant. A model can obtain high accuracy by predicting most pairs as negative, even if it fails to identify the rare positive interactions. For this reason, we report both Receiver Operating Characteristic - Area Under the Curve (ROC-AUC) and Average Precision (AP). Precision–recall based evaluation is particularly informative in settings where the positive class is rare, because it focuses directly on the trade-off between recovering positives (recall) and controlling false positives (precision) [17, 18].

ROC-AUC measures how well the model ranks positive interactions above negative interactions across all possible classification thresholds. It is useful as a general ranking metric, but can be overly optimistic in highly imbalanced settings because the number of true negatives is very large [17].

AP summarizes the precision–recall curve and focuses more directly on the model’s ability to accurately detect positive interactions. We therefore treat AP as the primary metric for model selection and comparison in the isoform–isoform link prediction task.

In addition to these ranking-based metrics, we also report threshold-based classification metrics using a probability threshold of 0.5. This yields a confusion matrix consisting of true positives, false positives, true negatives, and false negatives. From this matrix, we compute accuracy, precision, recall, and specificity. These metrics provide an interpretable summary of the model’s predictions at a fixed decision threshold, but they are interpreted as secondary to ROC-AUC and AP because the choice of threshold is not optimized, and is therefore arbitrary.

To assess whether observed performance differences between models are statistically significant, we perform comparisons across matched random seeds. Since the same 10 seeds are used for all models, pairwise comparisons are made between configurations with the same seed. We use the paired Wilcoxon signed-rank test, a non-parametric test that evaluates whether the median difference between paired observations differs from zero without assuming normality of the performance scores [19]. As models are tested

across multiple latent dimensions and modality configurations, this results in multiple pairwise tests, which increases the rate of type I errors. To control for this, we apply the Holm correction to the resulting p-values to control the family-wise error rate [20]. For comparisons against a fixed baseline, we use a one-sample Wilcoxon signed-rank test on the per-seed performance differences relative to the baseline model. Unless otherwise stated, statistical tests are performed on AP, and significance is assessed at the $\alpha = 0.05$ level.

Results are reported as means with 95% confidence intervals across random seeds. For a metric value x_s from seed s , the interval is computed as

$$\bar{x} \pm t_{0.975,9} \frac{\sigma_x}{\sqrt{10}},$$

where $\bar{x}$ is the sample mean across seeds, σ_x is the sample standard deviation across seeds, and $t_{0.975,9}$ is the 97.5th percentile of the Student t -distribution with nine degrees of freedom.

4 Results

This chapter reports model performance for the three experimental settings: transductive isoform–isoform prediction, inductive isoform–isoform prediction, and gene–gene reconstruction without direct gene–gene supervision.

4.1 Transductive isoform-isoform prediction

We first evaluate the LDM in the transductive setting.

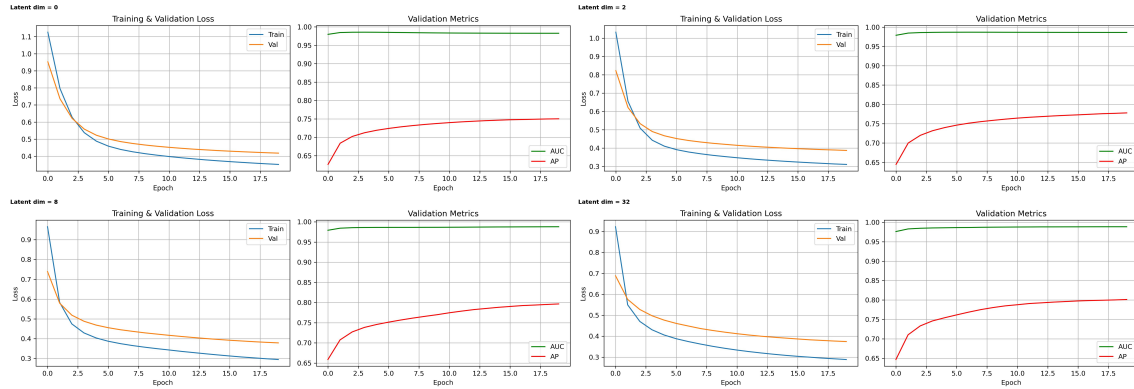


Figure 4.1: Training curves across latent dimension for seed 42. Left: train and validation loss. Right: validation AP and ROC-AUC

Figure 4.1 shows similar optimization behavior across dimensions. Training loss falls sharply in the first epochs and then declines more slowly without fully plateauing. Final loss values decrease slightly with larger latent dimensions. Validation ROC-AUC changes little during training, whereas validation AP improves gradually and reaches higher values for larger latent dimensions.

Table 4.1: Transductive isoform–isoform link prediction performance across latent dimensions.

Latent dimension d	AP	ROC-AUC	Δ AP vs. $d = 0$
0	0.747 ± 0.003	0.981 ± 0.001	—
2	0.774 ± 0.003	0.985 ± 0.001	$+0.028$ [0.027, 0.029]
8	0.791 ± 0.002	0.987 ± 0.001	$+0.044$ [0.043, 0.046]
32	0.795 ± 0.002	0.987 ± 0.001	$+0.048$ [0.047, 0.050]

Table 4.1 shows two effects. First, the random-effects-only model ($d = 0$) is already strong, with AP 0.747 ± 0.003 and ROC-AUC 0.981 ± 0.001 . However, latent geometry still improves performance. The AP gains Δ AP over the random-effects-only baseline are significant for all latent dimensions after Holm correction: Δ AP = 0.028 for $d = 2$, 0.044 for $d = 8$, and 0.048 for $d = 32$ (all $p_{\text{Holm}} = 0.006$). This also shows, that increasing latent dimension size yields gradually higher AP and AUC.

The result in the table is supported by Figure 4.2: larger latent dimensions improve precision across most recall values, with the strongest separation between $d = 0$ and the

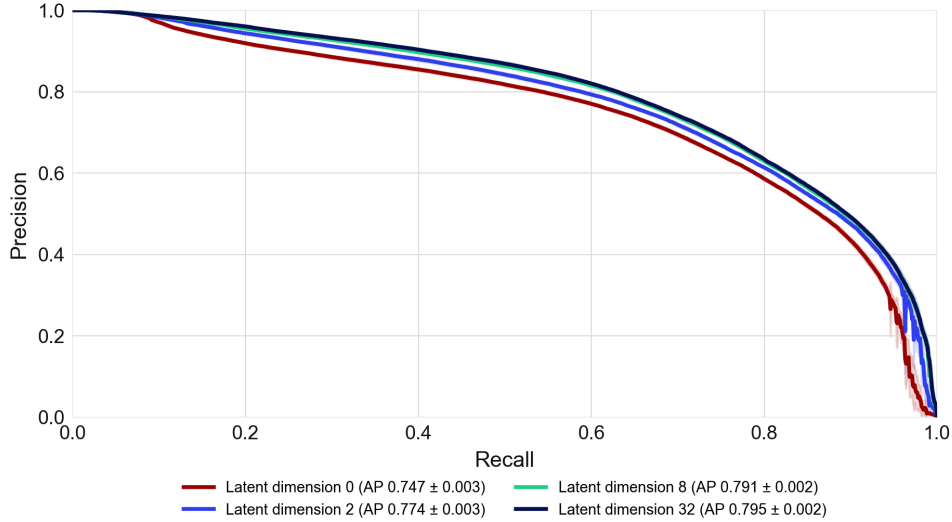


Figure 4.2: Precision–recall curves for transductive isoform–isoform link prediction across latent dimensions. Shaded regions indicate variability across random seeds.

models with latent geometry. The uncertainty in precision increases at very high recall, where retrieving the remaining positive examples requires accepting more false positives.

Figure 4.3 gives a thresholded view of the best transductive model ($d = 32$). Accuracy is high (97.2%), but this mostly reflects the many true negatives. The model achieves both high recall (90.8%) and high specificity (97.3%), indicating that it recovers most positive and negative interactions alike. Its main weakness is the low precision (38.6%), driven by the large number of false positives. This may stem from the positive class weight p_c in Equation (3.9), as well as from fixing the decision threshold arbitrarily at 0.5 rather than tuning it.

4.2 Inductive isoform-isoform prediction

We next evaluate the MMLDM in the inductive setting. In addition to overall test performance, we report AP separately for test pairs where only one isoform is unseen and for test pairs where both isoforms are unseen.

Figure 4.4 shows six representative training runs from seed 42: three modality configurations and the two extreme latent dimensions ($d = 0$ and $d = 32$). The main pattern is that the target isoform–isoform task changes little during training. $\mathcal{L}_{II}$ decreases only slightly, and validation AP and ROC-AUC are nearly flat. The auxiliary losses behave differently: $\mathcal{L}_{GI}$ is static, while $\mathcal{L}_{GG}$ decreases substantially when gene–gene supervision is included, especially at $d = 32$.

4.2.1 Modality ablations

We start by evaluating whether the auxiliary gene-level modalities improve inductive isoform–isoform prediction.

Table 4.2 shows no clear benefit from the auxiliary modalities. The II-only, II+GI, and II+GI+GG models have nearly identical AP across all dimensions. At $d = 32$, the full model improves over II-only by only 0.003 AP, and this difference is not significant ($p_{\text{Holm}} = 0.146$). The other modality comparisons at $d = 32$ are also non-significant: II+GI vs. II-only gives $\Delta\text{AP} = 0.001$ ($p_{\text{Holm}} = 0.432$), and adding G_{GG} on top of II+GI gives $\Delta\text{AP} = 0.002$ ($p_{\text{Holm}} = 0.211$).

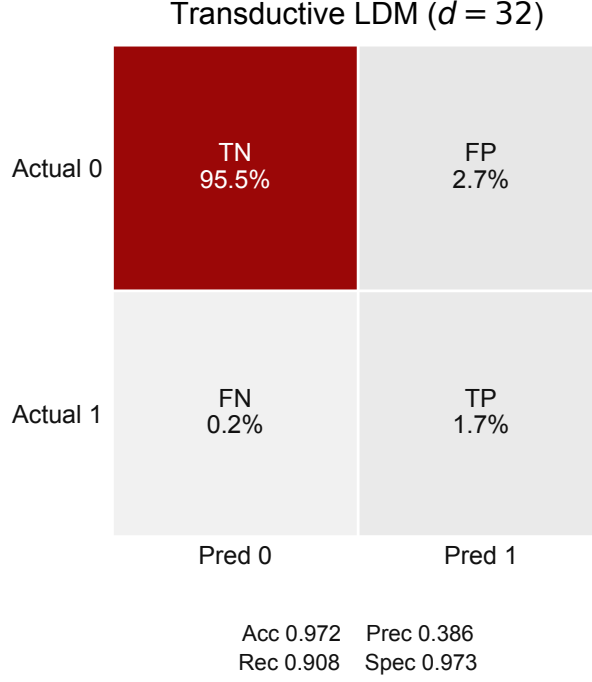


Figure 4.3: Mean normalized confusion matrix for the transductive LDM with $d = 32$ at score cutoff 0.5 across seeds 42–51.

Table 4.2: Modality ablation results for inductive isoform–isoform link prediction. The table reports overall AP for each modality configuration across latent dimensions.

Latent dimension d	II only	II + GI	II + GI + GG	Δ AP full vs. II only
0	0.492 ± 0.011	0.492 ± 0.012	0.492 ± 0.012	0.000
2	0.494 ± 0.012	0.495 ± 0.011	0.495 ± 0.011	+0.001
8	0.496 ± 0.011	0.495 ± 0.012	0.495 ± 0.012	−0.001
32	0.495 ± 0.012	0.496 ± 0.012	0.498 ± 0.012	+0.003

4.2.2 One-unseen and two-unseen test pairs

Despite no statistically significant difference in performance across modalities, we use the MMLDM with the full multimodal configuration in this section to investigate performance across latent dimensions and in the two separate interaction cases: interactions between two held-out isoforms vs. between one held-out and one observed.

Table 4.3 shows that inductive performance is much lower than transductive performance: the best full MMLDM reaches AP 0.498 ± 0.012 , compared with 0.795 ± 0.002 for the best transductive model. One-unseen pairs are consistently easier than two-unseen pairs, with AP differences from 0.039 to 0.046 across dimensions. These differences are significant for all dimensions after Holm correction ($p_{\text{Holm}} = 0.008$).

The effect of latent dimension is modest in the inductive setting. Compared with the $d = 0$ model, $d = 2$ and $d = 8$ do not significantly improve overall AP after Holm correction (Δ AP = 0.002, $p_{\text{Holm}} = 0.129$ and Δ AP = 0.002, $p_{\text{Holm}} = 0.232$, respectively). The $d = 32$ model shows a small but significant improvement over $d = 0$ (Δ AP = 0.006, 95% CI [0.003, 0.009], $p_{\text{Holm}} = 0.018$). The practical effect of latent geometry is therefore measurable but limited for inductive isoform–isoform prediction.

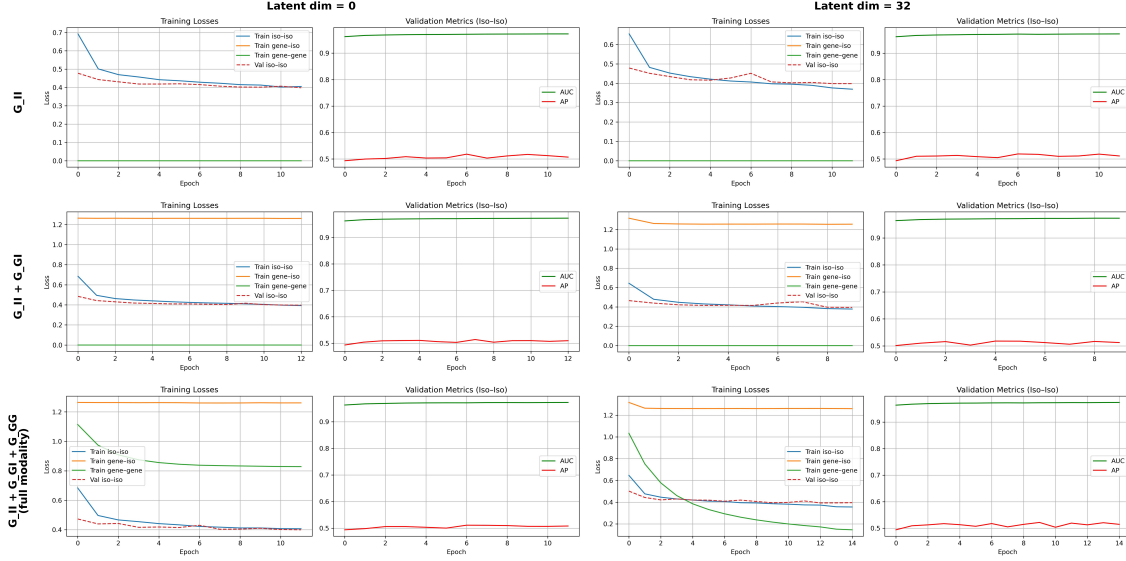


Figure 4.4: Training curves across modalities for latent dimension 0 and 32 (seed 42). Left: three train losses $\mathcal{L}_{II}$, $\mathcal{L}_{GI}$, $\mathcal{L}_{GG}$ and validation loss $\mathcal{L}_{II}$. Right: validation AP and ROC-AUC

Table 4.3: Inductive isoform–isoform link prediction performance for the full MMLDM.

Latent dimension d	Overall AP	ROC-AUC	One-unseen AP	Two-unseen AP
0	0.492 ± 0.012	0.970 ± 0.001	0.497 ± 0.011	0.458 ± 0.025
2	0.495 ± 0.011	0.971 ± 0.001	0.499 ± 0.010	0.458 ± 0.025
8	0.495 ± 0.012	0.971 ± 0.002	0.499 ± 0.011	0.461 ± 0.025
32	0.498 ± 0.012	0.972 ± 0.001	0.503 ± 0.011	0.457 ± 0.025

Figure 4.5 shows the precision–recall trade-off at a fixed threshold. Compared with the transductive model, false positives increase from 2.7% to 11.6% overall, while the true negatives drop from 95.5% to 86.7%. This indicates that the model overestimates the number of positive interactions, falsely labeling more negative interactions as positive. The effect is mirrored in the precision, which falls from 38.6% to 12.7%. However, the recall increases slightly from 90.8% to 94.6%, so the model captures a greater share of the true positive interactions, but at the expense of much lower precision. The two interaction types perform similarly in the thresholded setting: for interactions with one isoform observed during training, the model performs marginally better on every metric except the false negatives, which are equal.

4.3 Gene-gene reconstruction without direct supervision

Finally, we evaluate whether gene embeddings learned without direct gene–gene supervision can reconstruct gene-level structure in the STRING network. In this setup, the model is trained using the G_{II} and G_{GI} graphs, while the STRING-derived gene–gene graph G_{GG} is excluded from training. We evaluate gene interactions from two perspectives on the same STRING candidate pool. Distance scores a gene pair by $-\|u_g - u_h\|$ between learned gene embeddings. Aggregation scores it by the maximum isoform–isoform logit over all isoform pairs spanning the two genes. The $d = 0$ case is omitted for distance scores, since gene–gene reconstruction from latent distances is not defined without gene latent positions.

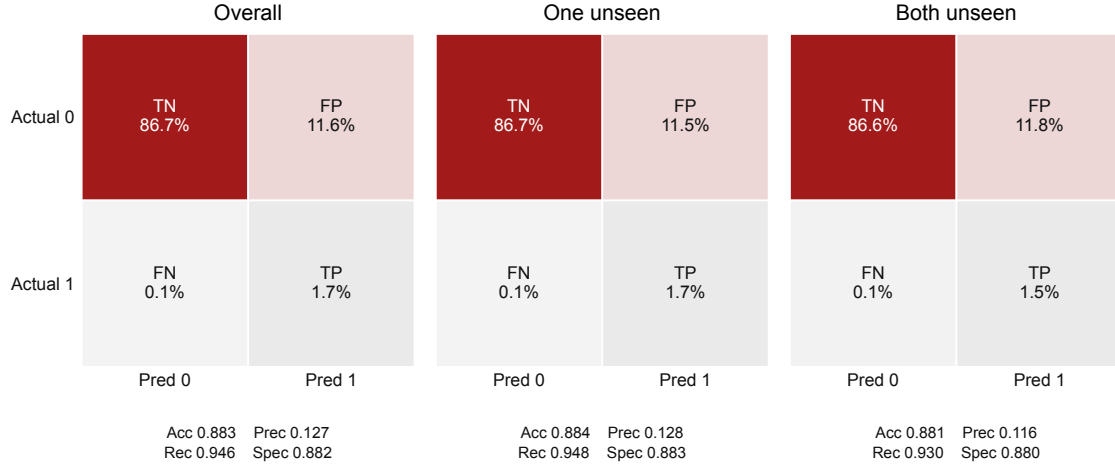


Figure 4.5: Mean normalized confusion matrices for the full inductive MMLDM with $d = 32$ at score cutoff 0.5 across seeds 42–51, split by overall, one-unseen, and both-unseen test pairs.

Table 4.4: Gene-level edge completion, comparing two readouts on the same candidate pool. The baseline AP is the STRING positive rate of the pool.

Dimension	Distance AP	Agg. AP	Distance ROC-AUC	Agg. ROC-AUC
Baseline	0.159	0.159	0.500	0.500
$d = 0$	—	0.272 ± 0.003	—	0.615 ± 0.001
$d = 2$	0.180 ± 0.002	0.272 ± 0.004	0.542 ± 0.003	0.615 ± 0.002
$d = 8$	0.190 ± 0.002	0.272 ± 0.003	0.552 ± 0.004	0.615 ± 0.002
$d = 32$	0.192 ± 0.002	0.271 ± 0.005	0.555 ± 0.003	0.614 ± 0.002

Table 4.4 shows that both readouts score above the STRING positive-rate baseline of 0.159. The distance readout improves with dimension, reaching AP 0.192 ± 0.002 and ROC-AUC 0.555 ± 0.003 at $d = 32$. This is a modest but significant AP gain over baseline ($\Delta\text{AP} = 0.033$, $p_{\text{Holm}} = 0.014$).

Unlike the distance readout, aggregation changes little with latent dimension. The aggregation AP is already 0.272 ± 0.003 at $d = 0$ and remains essentially unchanged for larger latent dimensions. Comparing $d = 32$ against $d = 0$ gives $\Delta\text{AP} = -0.002$ (95% CI $[-0.006, 0.002]$, $p_{\text{Holm}} = 1.000$). Because aggregation is already strongest at $d = 0$ and does not improve with d , the signal is likely dominated by isoform-level random effects rather than latent geometry.

Aggregation performs better than distance. At $d = 32$, it reaches AP 0.271 ± 0.005 and ROC-AUC 0.614 ± 0.002 , corresponding to $\Delta\text{AP} = 0.112$ over the baseline ($p_{\text{Holm}} = 0.014$). It also outperforms the distance readout at $d = 32$ by $\Delta\text{AP} = 0.079$ ($p_{\text{Holm}} = 0.006$).

Figure 4.6 shows the thresholded performance for the $d = 32$ model under the top-prevalence threshold. We use this threshold, since the 0.5 cutoff is meaningless in this setting. All distance scores are non-positive by definition, resulting in only negative predictions after applying sigmoid. This threshold instead predicts the same fraction of positives as the STRING positive rate. Aggregation recovers a larger fraction of positives than distance, increasing recall and precision from 0.206 to 0.302, while also improving specificity from 0.850 to 0.868. The confusion matrices therefore support the ranking metrics: the

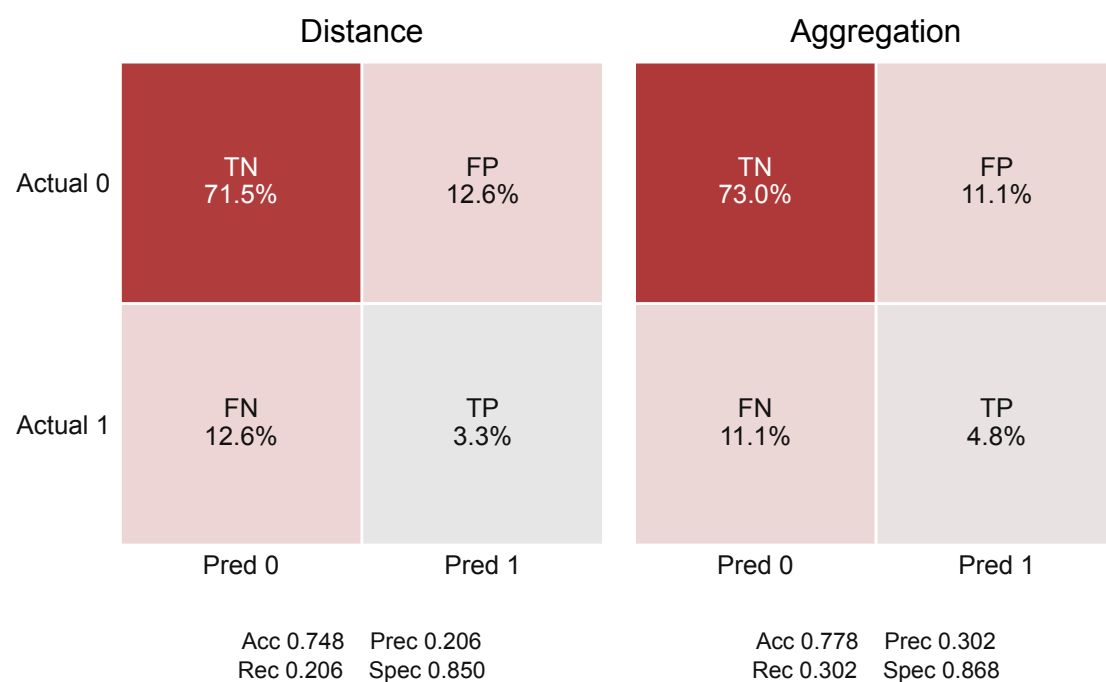


Figure 4.6: Mean normalized confusion matrices with top-prevalence threshold for the no-GG MMLDM with $d = 32$ on the STRING candidate pool across seeds 42–51.

max-aggregation readout captures more STRING-consistent gene-level structure than the direct gene-embedding distance readout.

5 Discussion

The purpose of this thesis was not only to obtain high link-prediction scores, but to examine what kind of structure can be learned from an inferred isoform-level interaction graph, and whether that structure can be transferred across biological resolutions. The three research questions therefore address different parts of the same problem. The transductive experiments investigate whether G_{II} contains latent dyadic structure beyond node-specific interaction tendency. The inductive experiments investigate whether this structure can be recovered for isoforms whose target interactions are absent from isoform-isoform training. The gene-gene reconstruction experiments then explore whether the shared latent space contains gene-level structure even when G_{GG} is not used directly as supervision.

A central point across all three experiments is that the target graph is an inferred interaction resource rather than a complete set of experimentally observed isoform interactions. The labels in G_{II} are derived from probabilities estimated from canonical-level physical interaction evidence, and the model is trained on a binary version of these probabilities. The results should therefore be interpreted as reconstruction and generalization within this inferred resource. They do not, by themselves, show that the highest-scoring model predictions correspond to experimentally verified isoform-specific interactions. This distinction is especially important because the biological motivation of the thesis concerns resolution: canonical interaction evidence is useful precisely because it is more available, but it can also hide isoform-specific variation. A model that performs well on the inferred graph may partly be recovering structure inherited from the canonical evidence used to generate it.

The thresholding of π affects this interpretation. By replacing the inferred probability with a binary label, the task removes information about the strength and uncertainty of the original inference. Pairs on opposite sides of $\pi = 0.5$ are treated as qualitatively different, while pairs far apart on the same side of the threshold are treated as equivalent. This makes the graph suitable for unweighted link prediction, but it also makes the task sensitive to uncertain cases near the threshold. In this sense, model error is not only a question of whether the LDM or MMLDM fails to learn a biological pattern; some errors may reflect the fact that a continuous inference problem has been converted into a difficult classification problem.

The restriction to labeled interaction records narrows the meaning of the reported performance. Although N_I contains 58,578 isoforms, the experiments do not rank all possible pairs in the complete $\binom{|N_I|}{2}$ space. They distinguish positive and negative labels within the available inferred records. This was necessary for computational reasons and was supported by preliminary experiments with implicit negatives, but it means that the evaluation is not equivalent to open-ended isoform interaction discovery across the entire interaction space. The AP values should therefore be read as performance on the inferred candidate set defined by the data, not as a direct estimate of performance on all possible isoform pairs.

5.1 Node-specific signal and latent geometry

The transductive results show that the LDM captures strong structure in G_{II} , but they also clarify what kind of structure dominates the task. The random-effects-only model

already performs well, with AP 0.747 ± 0.003 in Table 4.1. Since this model has $d = 0$, it has no latent distances and no dyadic geometry. Its performance must therefore come from the terms r_i and r_j , which encode how likely each isoform is to participate in positive interactions independently of its position in a latent space. This matches the structure of G_{II} described in Section 2.1: most isoforms have no positive interactions after thresholding, while a smaller set of isoforms accounts for the observed connected part of the graph. In such a setting, node-specific interaction tendency is already highly informative.

This does not make the latent distance component irrelevant. Adding $z_i \in \mathbb{R}^d$ improves AP consistently and significantly for all tested latent dimensions. The improvement from $d = 0$ to $d = 32$ is 0.048 AP, which indicates that G_{II} contains relational structure beyond what can be explained by node-specific effects alone. The latent space therefore captures pairwise organization: some isoforms are not only generally more interaction-prone, but are more likely to interact with particular regions of the graph. This is the part of the result that supports the use of a distance-based model rather than a pure degree-correction model.

The size of the improvement is nevertheless important. Most of the transductive performance is already present at $d = 0$, while the gains from increasing d are smaller and appear to diminish from $d = 8$ to $d = 32$. This suggests that the inferred graph contains geometric structure, but that it is secondary to the strong degree-related signal. In biological terms, the model is not primarily discovering a rich low-dimensional organization independent of node connectivity. It is first learning which isoforms tend to be involved in interactions at all, and then using latent distances to refine this with dyadic structure.

This interpretation also explains why AP is more informative than ROC-AUC in the transductive experiments. ROC-AUC is close to saturated for all dimensions, changing only slightly from $d = 0$ to $d = 32$. AP separates the models more clearly because it focuses on how well the rare positive interactions are ranked relative to false positives. The precision–recall curves in Figure 4.2 are therefore more useful for judging the added value of latent geometry than ROC-AUC alone. The same point applies to the confusion matrix in Figure 4.3: the fixed threshold gives an interpretable snapshot of the classifier, but the threshold is not optimized and the loss is class-weighted with positive class weight p_c . These factors highly influence the performance of thresholded interactions, so the threshold-based metrics should remain secondary to AP.

5.2 Inductive generalization and auxiliary modalities

The inductive setting changes the problem substantially. In the transductive LDM, each observed isoform can obtain its own learned z_i and r_i . In the MMLDM, isoform positions and random effects are instead produced from ESM C features through $g_\theta(f_i)$ and $h_\phi(f_i)$. This is a stricter learning domain: the model must learn a mapping from sequence-derived features to interaction behavior, rather than fitting free parameters for each isoform.

The lower inductive AP in Table 4.3 is therefore expected. The best full MMLDM reaches AP 0.498 ± 0.012 , compared with AP 0.795 ± 0.002 for the best transductive LDM. This drop should not be read simply as a weaker model. It reflects the difference between predicting held-out labels for mostly observed isoforms and predicting labels involving isoforms whose isoform–isoform interaction patterns are absent from training. The inductive result is therefore closer to the motivating use case of predicting interactions for newly characterized isoforms, but it is also a more difficult task.

The $d = 0$ inductive baseline is especially informative. Unlike the transductive $d = 0$

model, it does not learn one free random effect for each isoform. Instead, the model learns a projection $h_\phi(f_i)$, which maps the ESM C feature vectors to a node-specific random effect r_i^{II} . Its performance therefore suggests that ESM C embeddings contain information about general isoform interaction tendency. In other words, even without latent distances, the model can learn that some sequence-derived representations are associated with higher or lower probability of interaction. The small gain from $d = 32$ over $d = 0$ then suggests that sequence-derived latent geometry adds some pairwise information, but much less than in the transductive setting.

The difference between one-unseen and two-unseen test pairs supports the same interpretation. Performance is consistently higher when only one node is unseen than when both nodes are unseen. This is reasonable because one-unseen pairs retain a connection to the part of the isoform–isoform graph used to train the sequence projection. When both nodes are unseen, both representations must be learned purely through transfer from sequence-derived features and any auxiliary side information. The two-unseen subset is therefore more closely related to the pure inductive task that motivated the MMLDM, and the lower AP in this subset shows that this remains the harder part of the task.

The confusion matrices in Figure 4.5 support this observation with slightly better threshold-based performance for one-unseen interactions. Relative to the transductive model, recall increases by 3.8 percentage points while precision decreases by 25.9 percentage points, suggesting that positions produced from ESM C features preserve detection of true interactions but sharply reduce the model’s ability to suppress false positives.

The modality ablation results are the most important part of the second research question. The expectation behind the MMLDM was that G_{GI} and G_{GG} would provide useful biological context beyond sequence features alone. Under the current setup, this is not clearly supported. The G_{II} -only configuration performs almost identically to the richer configurations with no significant difference found. The largest full-model gain over G_{II} alone is only 0.003 AP at $d = 32$ in Table 4.2.

This does not imply that gene membership or canonical gene-level interactions are biologically irrelevant. A more careful interpretation is that the current MMLDM does not convert these modalities into improved isoform–isoform predictions. The structure of the auxiliary objectives helps explain why. G_{GI} is a deterministic membership graph: each isoform belongs to exactly one gene. This can align isoforms with their associated gene positions, but it does not contain information about isoform-specific interaction patterns. It may also be counterproductive for the target task if it pulls isoforms from the same gene toward a shared gene position and thereby reduces isoform-specific separation. G_{GG} contains richer relational information, but its influence on isoform representations is indirect. It shapes the gene positions u_g , and any effect on isoforms must pass through the shared geometry imposed by G_{GI} . If the gene–isoform objective provides only weak or static training signal, then improvements in the gene–gene objective may not propagate strongly to the isoform–isoform task.

The inductive training curves in Figure 4.4 support this reading. The gene–gene loss $\mathcal{L}_{GG}$ decreases when G_{GG} is included, especially at higher latent dimension, which shows that the model can learn structure in the auxiliary gene-level graph. At the same time, the gene–isoform loss $\mathcal{L}_{GI}$ remains completely static, showing no improvement during training. Furthermore, validation AP and ROC-AUC for the isoform–isoform task remain almost static, and early stopping is triggered in every run. The model is therefore learning something from G_{GG} , but that learned structure is not improving the target task in a measurable way. This distinction matters: the auxiliary modality is not ignored, but its

objective is not aligned strongly enough with held-out isoform–isoform prediction.

There is also a biological reason why this transfer may be difficult. The motivation for isoform-level modeling is that canonical-level interactions can hide differences between isoforms from the same gene. If this is true, then a gene-level interaction edge is only a coarse prior for isoform-level interactions. It may indicate that some product of two genes can interact, but it does not identify which isoforms carry that interaction. The weak modality effect may therefore reflect the same resolution gap that motivated the thesis in the first place. Gene-level information can organize the broader context, but it may be too coarse to substantially improve isoform-specific prediction.

5.3 Gene-level reconstruction and the direction of transfer

Although adding gene-level modalities does not clearly improve inductive isoform–isoform prediction, the model trained with $G_{II} + G_{GI}$ still retains gene-level structure that aligns with the external STRING gene–gene network, despite never receiving direct G_{GG} supervision. The results of reconstructing the gene network in this setting reveal that both readouts recover STRING structure modestly above the random baseline. The distance readout, built from the learned gene positions u_g , reaches 0.192 AP, whereas the aggregation readout recovers more of the structure with 0.272 AP.

This shows that the shared latent space can organize genes using only G_{II} and G_{GI} , but that this organization is only partially aligned with external gene–gene evidence. The model can therefore recover gene-level structure as an aggregation of isoform-level structure, but the modest performance improvement indicates that compressing isoform-specific patterns discards much of the signal.

The contrast between research question 2 and 3 suggests an asymmetry in the transfer direction. Moving from isoform-level interactions to gene-level organization is an aggregation problem (many-to-one): can multiple isoform interactions support a single gene-pair relation? Conversely, moving from gene-level interactions to isoform-level predictions is a disaggregation problem (one-to-many): can a single gene-pair relation support multiple particular isoform pairs? The results suggest that the MMLDM handles the aggregation direction more successfully than the disaggregation direction. The same asymmetry appears within research question 3 itself: the aggregation readout, which lifts isoform-level scores to the gene level, significantly outperforms the distance readout based on the compressed gene embeddings.

The performance of the two readouts across latent dimensions in Table 4.4 sharpens this interpretation. The distance readout improves significantly with latent dimension, indicating that the STRING-aligned signal it captures genuinely lives in the learned gene geometry: more latent structure lets the gene embeddings encode more information. In contrast, the aggregation readout shows no significant change with latent dimension and is already informative at $d = 0$, which indicates it is driven mainly by the isoform node-specific random effects. The two readouts thus capture STRING structure through different mechanisms: one geometric and dimension-dependent, the other dominated by node-level effects.

The gene–gene reconstruction result should still be interpreted carefully. Although the STRING gene–gene labels are external to the training graphs, they are not an independent source of biological information. G_{II} was itself inferred using canonical-level physical interaction evidence, and the STRING physical channel used here is derived from overlapping physical-interaction data. The different graph modalities are biologically related, but they are not completely independent sources of information, so even the above-baseline

performance partly reflects shared physical-interaction signal rather than fully external validation.

5.4 Limitations and future work

The main data limitation is the use of thresholded labels instead of the original inferred probabilities. A natural extension would be to model π directly, either as a soft target or through a loss that accounts for uncertainty in the inferred probabilities. This would avoid treating near-threshold pairs as certain positives or negatives and would better match the probabilistic nature of the data-generating process.

A second limitation is the treatment of negative examples. The experiments use the labeled records and do not train against the full set of implicit non-edges. This gives a tractable and controlled evaluation, but it does not test how well the model ranks positive interactions among all possible isoform pairs. Future work could use more systematic negative sampling such as candidate sets based on biological plausibility. This would make the evaluation closer to a practical discovery setting.

The MMLDM architecture also leaves room for improvement. The projection from $f_i \in \mathbb{R}^{1152}$ to $z_i \in \mathbb{R}^d$ is intentionally simple, and the ESM C embeddings are kept fixed. This supports interpretability and keeps the model close to the LDM framework, but it may limit how much sequence information is used. More expressive sequence projections, fine-tuning strategies, or graph neural network components could improve performance, although they would also reduce the simplicity and interpretability of the current model.

The weak modality effects suggest that future work should focus less on simply adding auxiliary losses and more on how information is transferred through biological resolutions. Instead of treating G_{GI} and G_{GG} as additional reconstruction objectives, gene-level information could be used as a prior on isoform positions, as a regularization term, or as a pre-training signal followed by isoform-level fine-tuning. The modality weights λ_{II} , λ_{GI} , and λ_{GG} were chosen manually, and a systematic search over weighting schedules could test whether the auxiliary objectives become useful under different balances. It would also be informative to evaluate modality effects on specific subsets of the graph, such as genes with many isoforms, isoforms from low-degree genes, or interactions involving isoforms with no positive training edges.

Finally, independent biological validation is needed before interpreting model predictions as candidate isoform-specific interactions. The strongest validation would compare high-scoring predicted links against experimental isoform interaction data or other independent biological evidence not used in the construction of G_{II} or G_{GG} . This would make it clearer whether the model is learning biologically meaningful interaction patterns, rather than only reproducing the structure of the inferred training resource.

6 Conclusion

Regarding the first research question, we found that a standard LDM provides strong transductive link prediction performance on the inferred isoform–isoform interaction graph, reaching an AP of 0.795 in a 32-dimensional latent space. Latent geometry consistently and significantly improved performance compared with a random-effects-only baseline. However, the strong performance of the baseline model (AP = 0.747) shows that much of the predictive signal is explained by node-specific interaction tendency rather than latent dyadic structure. The inferred interaction graph contains meaningful geometric organization, but degree heterogeneity remains a dominant source of predictive information.

For the second research question, we found that the MMLDM supports inductive prediction for isoforms whose interaction patterns are absent from the training graph, achieving a best AP of 0.498. However, performance was substantially lower than in the transductive setting, demonstrating the difficulty of generalizing from sequence-derived features alone. Furthermore, incorporating gene–isoform membership information and gene–gene interaction data produced only marginal and statistically insignificant improvements. These results suggest that while ESM-derived sequence representations contain useful information for predicting interactions involving unseen isoforms, the auxiliary gene-level modalities are too coarse to substantially improve isoform-specific prediction under the current modeling framework.

For the third research question, we found only limited evidence that gene-level interactions can be recovered from the learned embeddings. When trained without gene–gene supervision, the MMLDM reconstructed the held-out gene interaction graph only slightly above baseline. Using distances between the learned gene embeddings improved performance modestly, reaching an AP of 0.192 in the highest-dimensional latent space. Aggregating isoform-level interaction predictions produced better results, achieving an AP of 0.272 even when no latent geometry was included ($d = 0$). This mirrors the findings from RQ1, where random effects accounted for most of the predictive signal. Consequently, the observed gene-level reconstruction appears to come primarily from node-specific interaction tendencies inherited from the isoform layer rather than from meaningful geometric structure captured by the latent embeddings. While some gene-level information is present in the learned representations, the results suggest that the model learns only weak gene-level organization in the absence of direct gene–gene supervision.

Taken together, the results reveal an asymmetry in cross-resolution transfer. Information learned at the isoform level can be aggregated to recover broader gene-level organization, whereas gene-level information provides only limited benefit when transferred back to the more specific task of isoform interaction prediction. This suggests that the resolution gap between genes and isoforms remains a central challenge for computational modeling of protein interaction networks.

Finally, the interaction labels used throughout this work are inferred rather than experimentally verified. The models therefore demonstrate successful reconstruction and generalization within an inferred interaction resource, but not necessarily direct prediction of true biological interactions. Future work should explore probabilistic labels, improved negative sampling strategies, more expressive sequence models, and independent biological validation to determine how well these approaches generalize beyond the inferred network studied here.

Bibliography

- [1] Marc Vidal, Michael E. Cusick, and Albert-László Barabási. “Interactome Networks and Human Disease”. In: *Cell* 144.6 (2011), pp. 986–998. DOI: 10.1016/j.cell.2011.02.016.
- [2] Yu-Ting Tseng et al. “IIDB: A Database for Isoform–Isoform Interactions and Isoform Network Modules”. In: *BMC Genomics* 16.Suppl 2 (2015), S10. DOI: 10.1186/1471-2164-16-S2-S10.
- [3] Xinping Yang, Jasmin Coulombe-Huntington, Shuli Kang, et al. “Widespread Expansion of Protein Interaction Capabilities by Alternative Splicing”. In: *Cell* 164.4 (2016), pp. 805–817. DOI: 10.1016/j.cell.2016.01.029.
- [4] Mohamed Ali Ghadie et al. “Domain-Based Prediction of the Human Isoform Interactome Provides Insights into the Functional Impact of Alternative Splicing”. In: *PLOS Computational Biology* 13.8 (2017), e1005717. DOI: 10.1371/journal.pcbi.1005717.
- [5] Peter D. Hoff, Adrian E. Raftery, and Mark S. Handcock. “Latent Space Approaches to Social Network Analysis”. In: *Journal of the American Statistical Association* 97.460 (2002), pp. 1090–1098. DOI: 10.1198/016214502388618906.
- [6] Peter D. Hoff. “Bilinear Mixed-Effects Models for Dyadic Data”. In: *Journal of the American Statistical Association* 100.469 (2005), pp. 286–295. DOI: 10.1198/016214504000001015.
- [7] Nikolaos Nakis et al. “A Hierarchical Block Distance Model for Ultra Low-Dimensional Graph Representations”. In: *IEEE Transactions on Knowledge and Data Engineering* 36.4 (2024), pp. 1399–1412. DOI: 10.1109/TKDE.2023.3304344.
- [8] William L. Hamilton, Rex Ying, and Jure Leskovec. “Inductive Representation Learning on Large Graphs”. In: *Advances in Neural Information Processing Systems*. Vol. 30. 2017. DOI: 10.48550/arXiv.1706.02216.
- [9] EvolutionaryScale. *ESM Cambrian: Revealing the Mysteries of Proteins with Unsupervised Learning*. <https://www.evolutionaryscale.ai/blog/esm-cambrian>. 2024.
- [10] Shuo Zhang et al. “GraphIsoFun: A Graph Neural Network Based Approach for Splice Isoform Function Prediction”. In: *2021 IEEE International Conference on Bioinformatics and Biomedicine (BIBM)*. 2021, pp. 112–117. DOI: 10.1109/BIBM52615.2021.9669396.
- [11] Damian Szklarczyk et al. “The STRING Database in 2025: Protein Networks with Directionality of Regulation”. In: *Nucleic Acids Research* 53.D1 (2025), pp. D730–D737. DOI: 10.1093/nar/gkae1113.
- [12] Aric A. Hagberg, Daniel A. Schult, and Pieter J. Swart. “Exploring Network Structure, Dynamics, and Function using NetworkX”. In: *Proceedings of the 7th Python in Science Conference*. 2008, pp. 11–15.
- [13] PyTorch Contributors. *BCEWithLogitsLoss*. <https://docs.pytorch.org/docs/stable/generated/torch.nn.BCEWithLogitsLoss.html>. 2025.
- [14] Diederik P. Kingma and Jimmy Ba. “Adam: A Method for Stochastic Optimization”. In: *arXiv preprint arXiv:1412.6980* (2014). URL: <https://arxiv.org/abs/1412.6980>.
- [15] PyTorch Contributors. *ReduceLROnPlateau*. https://docs.pytorch.org/docs/2.12/generated/torch.optim.lr_scheduler.ReduceLROnPlateau.html. 2024.
- [16] Tobias Rodrigues Bjerre and William Kock-Andersen. *Characterizing Protein Interaction Networks using Graph Representation Learning: Project Repository*. <https://github.com/tob-euro/Protein-interaction-network>. 2026.

- [17] Jesse Davis and Mark Goadrich. "The Relationship Between Precision-Recall and ROC Curves". In: *Proceedings of the 23rd International Conference on Machine Learning*. 2006, pp. 233–240. DOI: 10.1145/1143844.1143874.
- [18] Takaya Saito and Marc Rehmsmeier. "The Precision-Recall Plot Is More Informative than the ROC Plot When Evaluating Binary Classifiers on Imbalanced Datasets". In: *PLOS ONE* 10.3 (2015), e0118432. DOI: 10.1371/journal.pone.0118432.
- [19] Frank Wilcoxon. "Individual Comparisons by Ranking Methods". In: *Biometrics Bulletin* 1.6 (1945), pp. 80–83. DOI: 10.2307/3001968.
- [20] Sture Holm. "A Simple Sequentially Rejective Multiple Test Procedure". In: *Scandinavian Journal of Statistics* 6.2 (1979), pp. 65–70. DOI: 10.2307/4615733.

Technical
University of
Denmark

Brovej, Building 118
2800 Kgs. Lyngby
Tlf. 4525 1700

www.compute.dtu.dk